

Skylnsight — Airline Passenger Satisfaction Analytics

Sector Name :Aviation / Airline Industry

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Context & Problem Statement

01

Sector Context

Highly competitive airline industry where customer satisfaction drives loyalty and repeat bookings.
Decision-makers: Management, CX teams, Marketing, Operations

02

Problem Statement

Airlines lack a clear, data-backed view of which segments and travel classes are experiencing weakest satisfaction — and how that connects to recommendation behaviour.

03

Objective

Identify key drivers of satisfaction to guide service improvements and investment decisions.

Data Engineering (Source to Sink)

01

Source

Kaggle - Airline Reviews Dataset, 22,329 reviews | 497 airlines | 6 columns

02

Cleaning

Handled missing values → replaced with "Unknown"
Converted ratings to numeric
Removed unnecessary columns
Standardised column names (lowercase, underscores) · Filled missing categoricals with "Unknown" · Converted recommended (yes/no text) → binary
is_recommended to fix a 100% recommendation bug · Converted overall_rating to numeric

03

Dictionary

Key Columns: airline_name, overall_rating, seat_type, type_of_traveller, is_recommended, rating_category

KPI & Metrics Framework

01

What are we measuring?

- Average Overall Rating — 3.14
- Median Rating — 1
- Recommendation Rate — 34.9%
- Low Rating Share — 68.3%
- High Rating Share — 21.1%

02

Why these KPIs?

- Average & Median Rating → directly measure passenger satisfaction level
- Recommendation Rate → shows if satisfaction converts to loyalty & word-of-mouth
- Low Rating Share → reveals scale of service recovery needed
- High Rating Share → sets the benchmark for what good experience looks like

Key Insights (EDA)

01

68.3% of reviews are Low-rated — dissatisfaction is not a small edge case, it's the majority

02

Median rating is 1 — more than half of reviewers gave the lowest possible scores

03

Economy Class dominates volume (18,853 reviews) but averages only 3.00 — highest commercial priority

04

Premium Economy averages 2.99 — almost identical to Economy despite higher price point

05

Only 34.9% recommend — customer advocacy is critically weak

03

Family Leisure (2.56) and Couple Leisure (2.74) are the weakest named traveller segments

Advanced Analysis



Complex Work Showcase

$r = 0.655$ — Rating strongly predicts recommendation behaviour
Welch t-test:
Recommended avg 5.72 vs Not recommended avg 1.76 — gap of 3.96 pts ($p < 0.001$, Cohen's $d = 1.82$) → difference is statistically proven, not random
Chi-Square:
Rating category ($V = 0.693$, strong)
· Traveller type ($V = 0.206$) · Seat type ($V = 0.170$) — all significant



New Understanding

One generic fix won't work. The data statistically confirms that different segments (seat class, traveller type) behave differently — which means the airline needs a targeted, segment-wise action plan, not a one-size-fits-all approach.

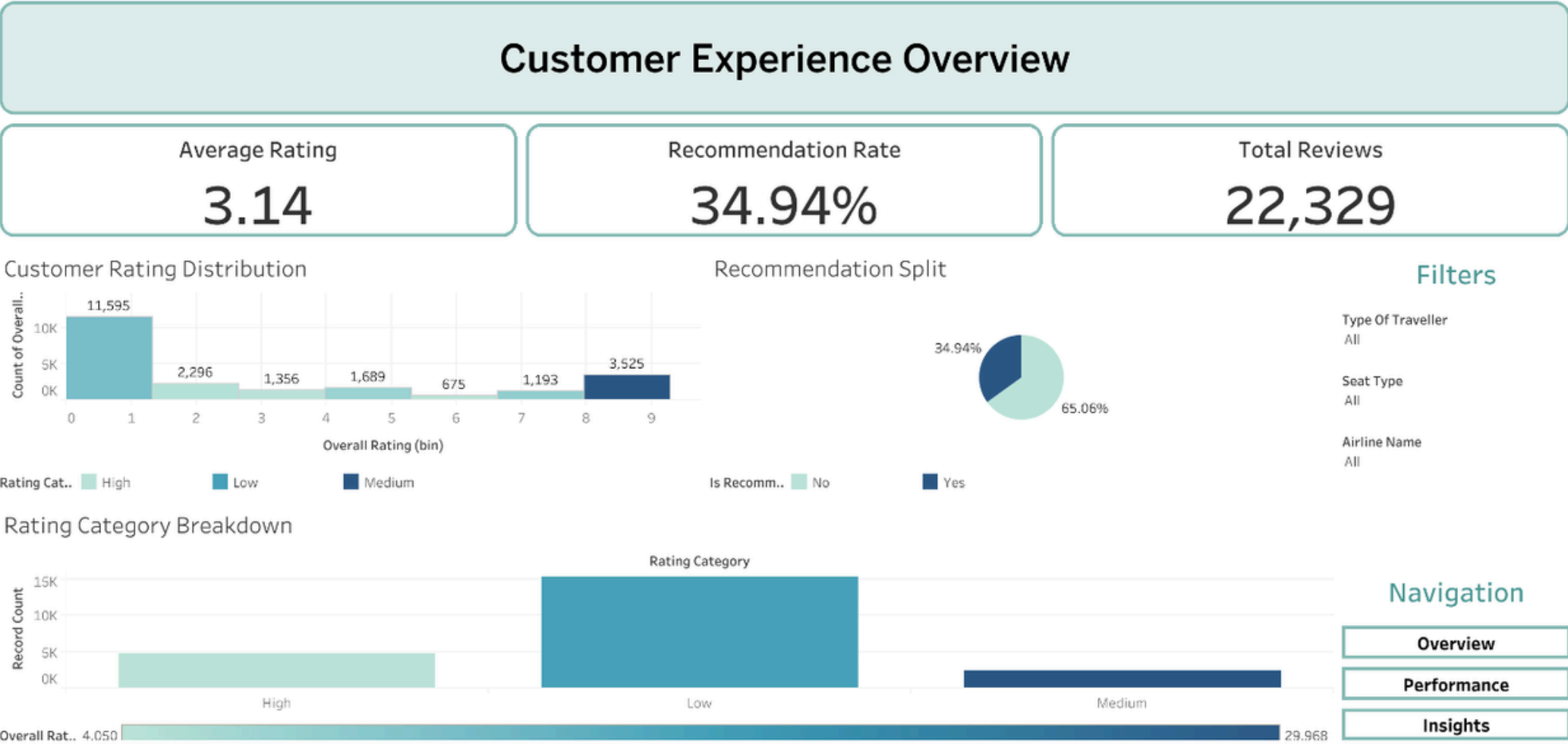




Dashboard Walkthrough

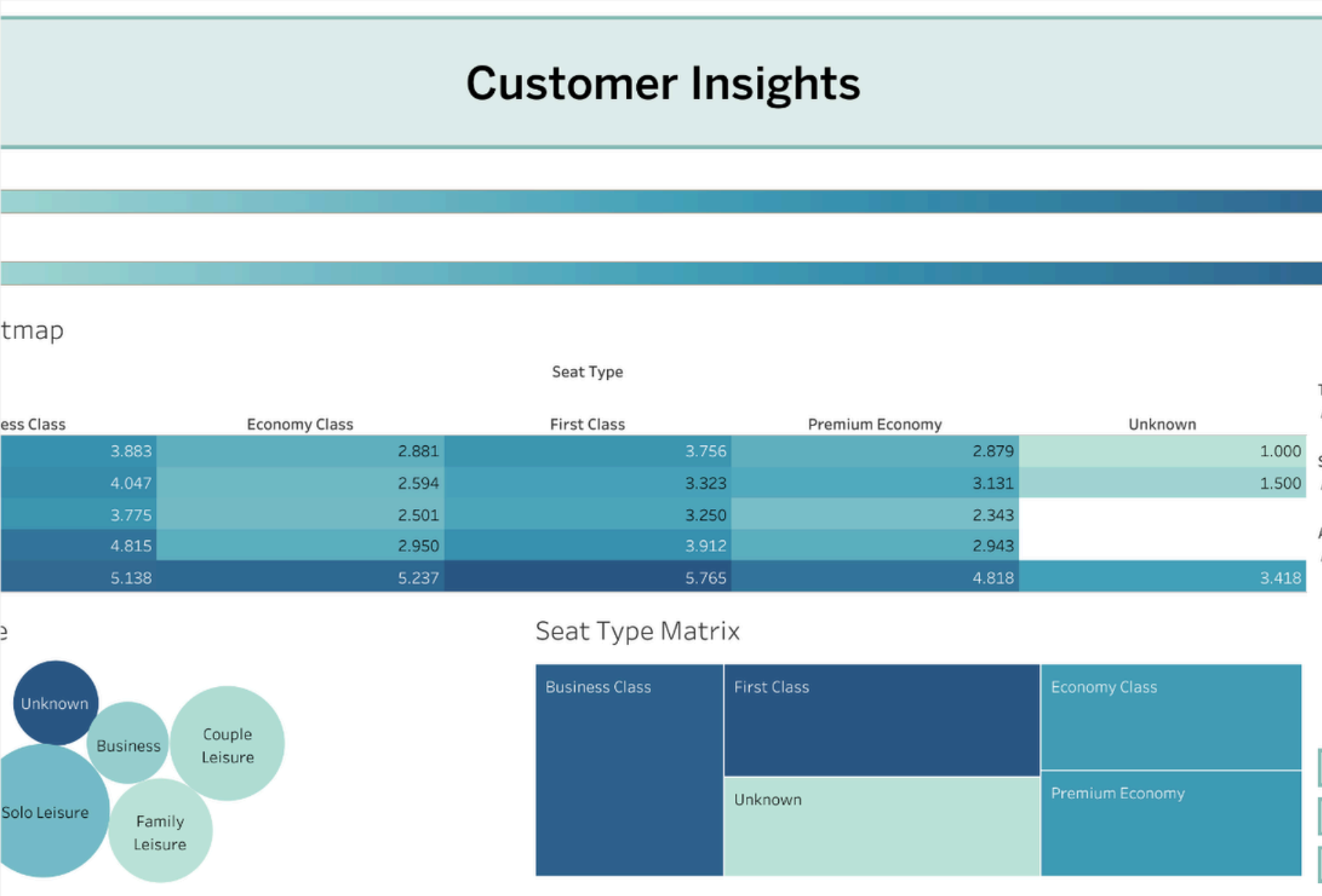
Executive View

3 KPI cards at top (Total Reviews, Airlines, Avg Rating, Recommendation Rate %) · Rating category distribution · Recommendation rate by rating category

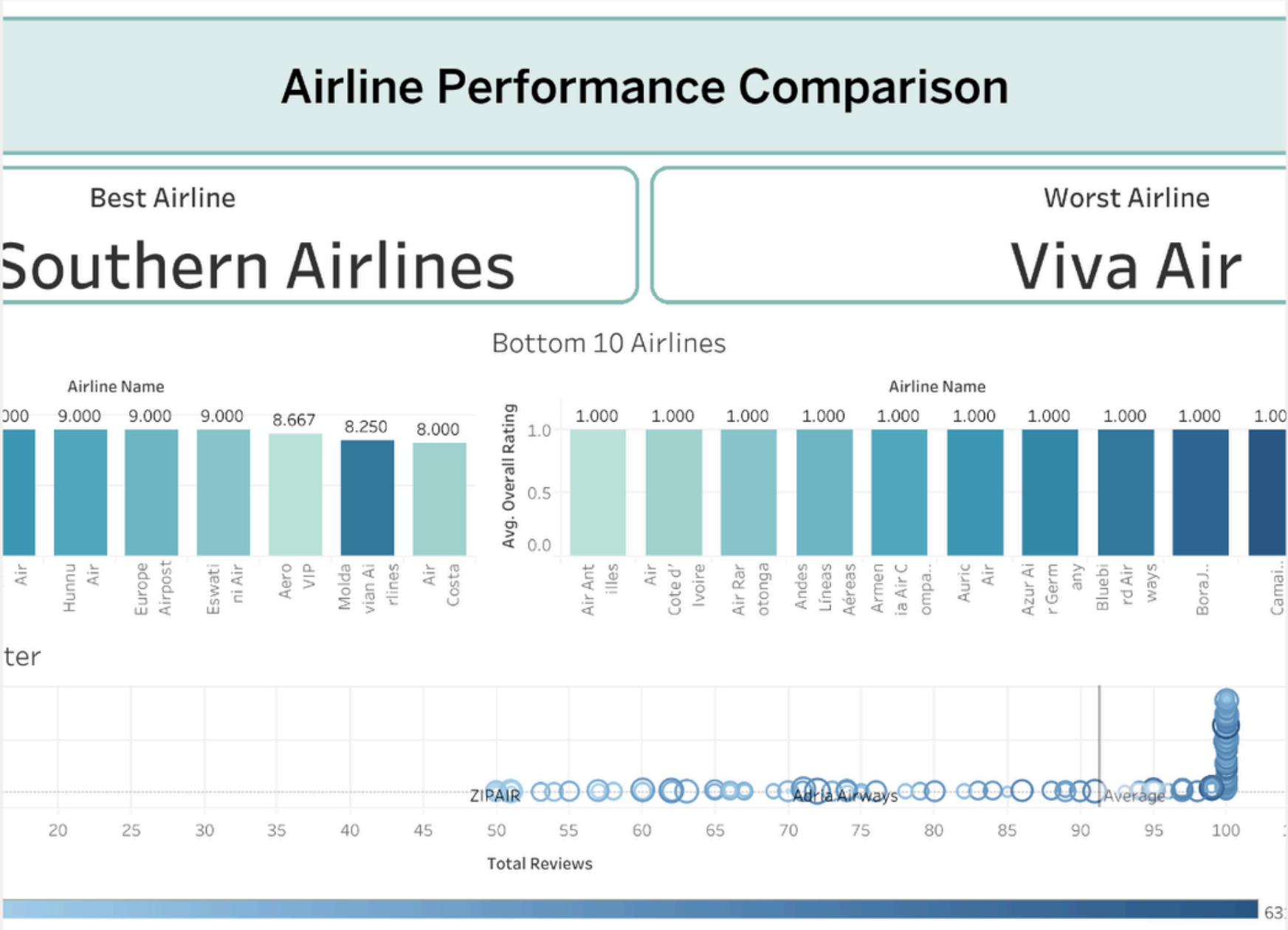


Operational View

Operational View: Avg rating by seat type and traveller type .
Top 10 / Bottom 10 airline benchmark charts . Filters for airline, seat type, traveller type, rating category, recommendation flag



China Southern Airlines leads the rankings while Viva Air ranks lowest, showing a sharp gap in customer satisfaction across airlines.



Recommendations

01 Launch Economy & Premium Economy recovery programme — target the most common complaint drivers in the highest-volume segment

03 Create a leisure-family service playbook — cover seating together, baggage clarity, disruption communication for Family and Couple Leisure travellers

02 Add KPI validation checklist — prevents errors like the 100% recommendation bug before dashboard publication

04 Benchmark top airlines — convert Myanmar Airways, Druk Air, China Southern's strengths into a practical checklist for weaker airlines

Impact & Value

01

Estimated Impact

+5pp recommendation rate → ~1,116 more recommending passengers
+8pp recommendation rate → ~1,786 more recommending passengers
Low rating share 68.3% → 60% → ~1,850 reviews exit Low category
KPI validation added → prevents wrong executive decisions from data bugs

02

Stakeholder Approval

68.3% reviews are Low-rated & growing in public channels. This dashboard gives leadership a measurable way to track improvements — act now before dissatisfaction compounds further.

Limitations & Next Steps

01

Data Gaps and Biases

- Final CSV has only 6 columns — no service-level root cause possible
- No date fields — trend/seasonality analysis not possible
- No review text — sentiment analysis not possible
- Self-selected public reviews — biased toward extreme experiences

02

Future Improvements

- Add service-rating columns (comfort, food, staff, entertainment) to ETL
- Include review_date for time-series and seasonal analysis
- NLP on review text for complaint-driver extraction
- Build predictive model for recommendation probability